EFFECTIVE RELATIVE CYCLES ARE EQUIDIMENSIONAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Closedness of the support of effective relative cycles (relative-cycles chapter, Lemma lemma-effective-support-closed, canonical tag 0H5W). cited at proof lines 1988-1989. Supp(alpha) is closed in X when alpha is effective.

Source reference. Original lemma 0H5W: sources/stacks-chow/tags.txt, source lines 21401–21401. Cited at proof lines 1988-1989: Supp(alpha) is closed in X.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Assume S locally Noetherian and f locally of finite type. Let $r, e \geq 0$ be integers. Let α be a relative r-cycle on X/S. If α is effective, then α is equidimensional.*

Proof. Assume α is effective. By Lemma 0H5W the support $\mathrm{Supp}(\alpha)$ is closed in X. Thus α is equidimensional as the fibres of $\mathrm{Supp}(\alpha) \to S$ are the supports of the cycles α_s and hence have dimension r. □

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